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Vikas **Raikwar**

··Instrumentation and Control Engineer··

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Instrumentation, Automation & Embedded Systems Engineer with 6.5+ years of hands-on experience in automation-driven food,dairy,assembly & production line processing plants. Strong expertise in PLC programming, firmware development, industrial communication, hardware debugging, and embedded systems (AVR, Arduino, Raspberry Pi). Proven experience in interfacing VFDs, Soft Starters, Servo Drives, HMIs, and industrial control panels, along with advanced exposure to AI-integrated embedded solutions. Adept at commissioning, troubleshooting, and maintaining mission-critical industrial automation systems.

CORE SKILLS

- PLC Programming & Automation
- Industrial Drive & Motion Control Integration
- Hardware Debugging & Field Troubleshooting
- Industrial Communication Protocols
- SCADA / HMI Development
- Control Panel Commissioning
- Embedded Firmware Development (Embedded C, Arduino, AVR, Raspberry Pi)
- AI-Integrated Embedded Systems
- Pneumatic Systems

WORK EXPERIENCE

Instrumentation & Control Engineer

Sanchi Milk Products (BKDS), Madhya Pradesh

Experience: 6.5+ Years | Present

- Responsible for end-to-end automation and instrumentation of dairy and food processing plants.
- PLC programming, commissioning, and maintenance of fully automated production lines.
- Interfaced VFDs, Soft Starters, Servo Drives, and HMIs for conveyors, processing machines, pumps, and mixers.
- Designed, debugged, and maintained industrial control panels (hardware + software).
- Worked on RS232, RS422, RS485, Modbus RTU communication for industrial devices and instruments.
- Performed hardware debugging and field troubleshooting using multimeters, wiring analysis, and signal checks.
- Coordinated with production and maintenance teams to minimize downtime and improve system reliability.

PLC, DRIVES & INDUSTRIAL INTERFACING

PLC & Industrial Device Interfacing

- Interfaced and commissioned VFDs, Soft Starters, Servo Drives, and HMIs with PLC systems for industrial automation.
- Worked on control panels including wiring, commissioning, fault diagnosis, and optimization.
- Hands-on experience with industrial motion control, speed control, torque control, and process automation.
- Integrated and tested industrial sensors (pressure, temperature, proximity, level, flow, encoders).

Industrial Communication

- Worked extensively on: RS232 / RS422 / RS485 ,Modbus RTU,IOT Gateways,Level Transmitters,RTD,Thermocouple,Encoders
- PLC-to-drive, PLC-to-HMI, and PLC-to-instrument communication setup and troubleshooting.
- Debugging of communication faults at hardware and protocol level

EMBEDDED SYSTEMS & AI EXPERIENCE

- Advanced programming on AVR microcontrollers, Arduino platforms, and Raspberry Pi.
- Firmware development using Embedded C, C++, and Python.
- Developed AI-integrated embedded solutions for monitoring, diagnostics, and automation logic.
- Designed and implemented embedded projects involving sensors, motors, wireless modules, and control algorithms.
- Experience in real-time data acquisition, processing, and intelligent decision-making at edge level.

INDUSTRIAL AUTOMATION STACK

- **PLCs:** Siemens, Delta, Mitsubishi, Innovance
- **SCADA/HMI:** Siemens TIA Portal ,WinCC
- **Drives:** VFDs, Soft Starters, Servo Drives
- **Protocols:** Modbus RTU, RS232, RS422, RS485,OPC UA/DA
- **Tools:** TIA Portal, GX Works, WPLSoft,ISPSoft, AutoShop (Innovance)

ACADEMIC ACHIEVEMENTS

- **Rank 1,** World Skills 2021 – MP Skills under Mechatronics trade at Symbiosis University, Indore (M.P.)
- **Rank 1**, World Skills 2018 – MP Skills under Mechatronics trade at Global Park, Bhopal (M.P.)
- **Rank 1**, Project Exhibition at Horizon Techfest 2015-16 B.T. Institute of Research & Technology 4THSemester, B.E.
- **Rank 1**, Robo-Race at Horizon Techfest 2015-16 B.T. Institute of Research & Technology 4THSemester, B.E.
- **Rank 2**, Robo-Race at Horizon Techfest 2016-17 B.T. Institute of Research & Technology 6THSemester, B.E.
- **Rank 2**, Robo-Race at Prabhav Techfest 2016-17 Infinity Institute of Research & Technology 7THSemester, B.E.

CO-CURRICULAR ACTIVITIES

- **WORKSHOPS :** Ethical Hacking, Quadcopters, Robotics, Wireless Robots, MATLAB, Line & Maze follower robot.
- **CERTIFICATIONS:** Interfacing of Microcontrollers at ROBOTRONIX, Jabalpur [M.P.], Wireless Robotics at ROBOTRONIX, DAVV Indore [M.P.], Basic Telecom Training at BRBRAITT,Jabalpur [M.P.], Quadcopters at IIT Roorkee

PROJECT/TRAINING/SOFTWARE PROFICIENCY

MAJOR PROJECT :
(Robotic Arm with teach-in function)

Robotic Arm implemented using ATmega328p micro-controller ,micro servo motor(SG90) & 3D printed parts .Technically based on Analog value control which are provided as input for arm movement in 4DOF .Consist of teaching algorithm which records analog value in array in record mode & executes movement of robotic arm for infinite loop when play-mode is active.

MINOR PROJECT:
Automatic rotating solar panel with self calibration system

Automatic rotating solar panel designed for automatic rotating towards high photonic rays for getting maximum efficiency from sun rays and transform into electrical energy by using solar panel. It is sensor less rotating solar panel which works on principle of maximum efficiency ,it automatically detects and calibrate itself to the direction getting maximum energy with help of servo motor(SG-90).

QUADCOPTER:

PID controlled self level DRONE which can lift 1.5kg weight and can be controlled through transmitter within range of 500m (Dia.) video recording and other application such as monitoring and surveillance

can be done. Arduino Nano & MPU6050 Gyro was used for making flight controller with Bluetooth (HC-05) and FLY Sky Transceiver system which works on frequency range of 2.4 GHz.

OTHER PROJECTS : 3D Printer, Wireless Bluetooth Controlled BOT, GSM Based Automatic Distribution Pole Cut-off with Temperature Monitoring System,DTMF robot, Automatic soil-moisture control system with moisture sensor.

BASIC TELECOM TRAINING :
BRBRAITT -BSNL
Jabalpur (M.P.)

Basic Telecom. Training in which overview of Telecom & mobile network, losses in wired & wireless networks, optical fiber communication with transmission &reception, mobile technologies & its generations, FTTH technology with access& transmission networks, basic principles of electronic changes, next generation networks and detail information about internet & IP addresses were discussed & practical information was provided during training session.

SOFTWARE PROFICIENCY : Express PCB,Proteus,MultiSim(Basics) ,Arduino IDE,Netbeans,,Fritzing PCB, MATLAB(Basics), Microsoft Word/Excel/PPT, AutoCAD 2010 (Basics),NetBeans 6.0(Basics) ,TIA PORTAL,WPL soft,GX Works,Autoshop(Innovance),FluidSIM,Simatic Manager 7,WPLsoft,Inkscape,Fusion 360,Solidworks 2019(Basics).

PROGRAMMING LANGUAGES: C++,Embedded C(Basic),JAVA,HTML(Basics), PYTHON(Intermediate) ,Ladder.

OTHER AUTOMATION

INDUSTRIAL AUTOMATION : Have experience in Siemens /Delta/Mitsubishi/Innovance PLC programming for automating industrial devices and equipment such as elevator ,conveyors, processing machines as well as commissioning of various industrial sensors.

Professionally Worked & Attended Service calls with:

- HERSHEYS CHOCOLATE PRODUCTS
- TREELY AGRO [Particle Board Manufacturing Industry]
- SANCHI MILK PRODUCTS

HOME AUTOMATION: Have experience in Home Automation with IOT & microcontrollers programming for automating Home equipment's such as fan ,switches, surveillance & monitoring with embedded technology sensors .

ACHIEVEMENTS EXTRA-CURRICULAR

RoboWar Co-ordinator, at Horizon Techfest 2016 BTIRT	-	Assisted to manage the Robowar field and verifying Robots per allotted criteria.
Badge Holder at Soft skill Development Session, BTIRT	-	Served as group leader for maintaining discipline and promoting importance of communication skills.
Co-ordinator of Online Exam, BTIRT	-	Assisted to look after students occurring in exam and give information regarding online exam software.
Honoured & awarded as Super Student,BTIRT	-	In recognition of loyality,leadership,sincerity & determination &Displayed during soft skills development program.
Workshop Organizer at BTIRT ,Sagar	-	Organized workshop on wireless Bluetooth robotics at BTIRT campus, Sagar (M.P.)
Workshop Organizer at Shailesh Memorial School,Sagar	-	Organized workshop on Basic Electronics for 9 -12 standard students.
Represented Madhya Pradesh at World Skill India 2018	-	Under Mechatronics skill, represented M.P. State in regional competition at World skill India 2018 held at JECC,Jaipur (Rajasthan).
Technical Co-ordinator, at technical workshop,BTIRT	-	Exhibited skills in software related queries,component monitoring & PCB designing at basic electronics workshop at BTIRT campus,Sagar(M.P.)

Honoured as Official Web Developer,BTIRT	-	Official web developer for www.satishb3anand.com under supervision of International author- Satish B3 Anand

AREAS OF INTEREST

Embedded systems	-	Making projects with microcontrollers such as 8051,AVR,Arduino with implementing various modules &sensors like Temperature,accelerometer,gyro,Proximity,Infrared sensors & GSM module and many other sensors with digital circuits
Robotics & UAV's	-	Making bots with various functions like gesture controlled,light & line following,maze solving bot and UAV's like Quadcopter,airplane with wireless protocols such as wifi,2.4ghz RF,Bluetooth.
PLC & Automation	-	Skilled in developing logic for automating and controlling industrial equipment and processes. Experienced in integrating and configuring a wide range of VFD & Soft-starters , industrial sensors, including reflectometers, inductive sensors, and color sensors,wireless and Serial communication with RS232 ,RS485,RS422 to enhance system functionality and efficiency.

DECLARATION

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.

Date & Place : _____ Vikas Raikwar